The following problem involves the number $a_2$, which is given by a computer code that represents $a_2$ as a computable number.

The code for $a_2$ is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
       k=k+1
       sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and $a_2$ per second, respectively. The treatment times we call $x_1$ and $x_2$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $1$.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the $\ell^{\infty}$ norm, if $a_2$ is given as in the codes above. Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 163 seconds

The number $a_2$ is given by the limit as $n \to \infty$ of Compute_a_2($n$). Evaluating the code:

- Compute_f($n$) initializes $\mathrm{sum} = 72^{-N}$ with $N = 10^{24}$, then for $k = 3$ to $n + 2$ adds $(1/2)^k - 72^{-Nk}$.
  As $n \to \infty$, the sum of $(1/2)^k$ is $1/4$, the sum of $72^{-Nk}$ is astronomically smaller

than $72^{-1N}$, so $f(\infty) = \frac{1}{4} + \delta$ with $\delta > 0$ extraordinarily small.

- Then Compute_a_2($n$) = $2f(n) + \frac{1}{2} + 0.25^n$.
  In the limit, $2f(\infty) = \frac{1}{2} + 2\delta$ and $0.25^n \to 0$, giving $a_2 = 1 + 2\delta > 1$.

The optimization problem: minimize $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$.
Since $a_2 > 1$, the objective $T = 1 + (1 - a_2)x_2$ is minimized by taking the largest possible $x_2$. The constraint $x_1 = 1 - a_2 x_2 \geq 0$ gives $x_2 \leq 1/a_2$. The optimum is $x_1 = 0$, $x_2 = 1/a_2$.

Because $a_2 = 1 + (\text{tiniest positive number})$, $1/a_2$ is essentially $1.000\ldots$ to far more than two decimal places. To two-digit accuracy in the $\ell^\infty$ norm:

$$x_1 = 0.00, \quad x_2 = 1.00$$